Introduction to sampling

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Course overview

1. Population, sampling, sample
2. Sampling designs
3. Estimation
4. Multistage sampling
5. Multiphase sampling
6. Recap: main formulas
7. References

Population, sampling, sample

Why sample?

What is sampling?

Population, sampling, sample

Why sample?

The question does not arise when we can measure everything, but...
Using partial data has long been controversial!

What is sampling?

Sampling makes it possible to study a population by examining only part of it.

Population, sampling, sample

Clearly defining the population

Statistical population under study

➔ areas and trees.

NB

- A fraction of the population may **not** be sampleable (inaccessible, dangerous, or prohibited areas).

- the population may be infinite

- units may have different sizes, but not different natures

Example: McRoberts et al.

Sampling designs for national forest assessments

Ronald E. McRoberts,[1] Erkki O. Tomppo,[2] and Raymond L. Czaplewski[3]

1.3 Defining the population

Scientifically defensible estimation of population attributes is based on a formal body of mathematical theory, which must be respected if it is to be used to defend the accuracy of sample-based estimates. The careful selection of a sampling frame, plot configuration and sampling design are crucial steps in the process and cannot be accomplished independently of each other. Each decision has an impact on the others.

The mathematical theory begins with a precise definition of the population for which attributes will be estimated. For example, for a municipality of 5 million ha of which 1 million ha comprises forest, the statistical population could be described in several different but logical ways:

- Thousands of tree-stands and non-forest polygons
- Tens of millions of potential 0.1 ha sampling plots
- Ten million remotely sensed 30 m x 30 m pixels
- Billions of trees
- An infinite number of points.

Population, sampling, sample

Sample

A miniature of the population:

sampling makes it possible to study a population by examining only part of it.

Population, sampling, sample

Measuring all trees would be impossible (and unnecessary).
Statistical sampling means choosing a **sampling design**
to select a fraction of these trees
and draw conclusions about the entire tree population.

➔ the choice will be very important

Population, sampling, sample

Probability sampling is contrasted with **non-probability** (empirical) sampling.

Probability sampling is based on probabilities,
the probabilities of selecting an individual (observation unit).

Population, sampling, sample

In a population U, we study the values (characteristics) x et y measured on observation unit k.

Let:

U	the population of size N
X, Y	a population characteristic to be determined
x, y	a characteristic of the observation unit k

we seek to estimate

$$X = \sum_{k \in U} x_k, \; Y = \sum_{k \in U} y_k$$

ou

$$\bar{X} = \frac{1}{N} \sum_{k \in U} x_k, \; \bar{Y} = \frac{1}{N} \sum_{k \in U} y_k$$

Population, sampling, sample

Randomness lies exclusively in the selection of units

Let U be a population

$$U = \{1, ..., k, ..., N\}$$

A sample S is drawn from U.

We are interested in the variable of interest y
which takes the value y_k for individual k of U.

In sampling theory, randomness concerns the selection of observation units (i.e. the k), not their values themselves (i.e. not the y_k).

Population, sampling, sample

Attention

In sampling theory, x_k and y_k are considered non-random, and the only source of randomness is whether or not they are observed.

To understand this, consider the example of a dice.

Population, sampling, sample

There are 6 values

The values on the faces are fixed.

Only the outcome of the draw is random, not the values on the faces of the dice.

Example: a dice roll

1	2	3	4	5	6
$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

Population, sampling, sample

Definition

Sample = a subset of the population

➔ Resulting from a choice (and therefore from a sampling procedure or design)

Choice result no. 1 = list of selected units

$$a = (a_1, ..., a_N)^t$$

A sampling design without replacement is a *probability distribution*.

Definition

The list of all elements making up the population is called **sampling frame** (Dice example: the six faces)

Population, sampling, sample

Other common definitions:

sample : set of units selected from the population

random sample : set of units selected from the population ensuring a known and non-zero probability in advance of belonging to the sample

Population, sampling, sample

Probability or **random** sampling

each individual in the population has a **known and non-zero probability in advance** of belonging to the sample.

This probability is the **selection probability** of an individual in the sample.

In non-probability sampling, the inclusion probability of individuals is not known.

Population, sampling, sample

A sampling design without replacement is a probability distribution.

Let s be a sample drawn from population P,

$$p(s) \geq 0 \text{ for all } S \text{ in } P$$

$$\sum_{s} p(s) = 1$$

Dice example

Each face has a one-in-six chance of being rolled

$$p = 1/6$$

$$\sum_{s=1}^{6} p(s) = 6 1/6 = 1$$

Population, sampling, sample

- **Random sample** : each sample S has a probability $P(S)$ of being selected
- An estimator is a function of the x_i for $i \in S$
- a characteristic θ $(x_i, i \in U)$ is estimated by $\hat{\theta}(x_i, i \in U)$
- Si S is a random sample, then $\hat{\theta}$ is a random variable for which its expectation and variance can be calculated.

Population, sampling, sample

Soient

- A population U de taille N;
- S is a **random sample** of size n;
- Y a variable of interest, observed on units 1 to n;

Alors

$$S \subset U$$

$$n \leq N$$

$f = \frac{n}{N}$ is the **sampling fraction** of the survey.

Sampling designs

Definition

Sampling design = method or procedure used to select the sample.

NB

Not every sampling design ensures equal-probability selection of units.
A sampling design is associated with one or more estimation methods

The most common **probability sampling designs** are:

- random
- systematic
- stratified

Simple random sampling design

Simple random sampling is obtained by randomly selecting the elements to be measured.

Reminder:

Sampling frame Complete list of all elements of the population of size N

Sample subset of the selected population

Sampling unit Basic unit of the survey, often the individual



Systematic sampling design

Systematic sampling requires determining:

- the interval (distance) between two sampled points along both axes
- the position of one of the points, e.g. the central point, from which distances are calculated along both axes.



Sampling designs

Stratified sampling design



Sampling designs

What is stratification?

Stratification consists of creating homogeneous groups of sampling units

- Basic idea: minimize within-group variance (= maximize within-group similarity)
- Principle: partition into homogeneous sets

Sampling designs

Stratification reduces variance. Extreme hypothetical example:

groups	values	mean	variance
total	10,10,10,10,10,10,20,20,20,20,20,20	15	5,22
group 1	10,10,10,10,10,10	10	0
group 2	20,20,20,20,20,20	20	0

Sampling designs

Attention

Stratification uses information external to the sampling process (to create homogeneous groups).

If the information used for the partition is obtained through the sample, then the stratification is de facto a **post-stratification**

Randomness!

Proportions of the colors in an urn

Urn contents

40 red balls
60 black balls

Random sampling with replacement:

- 2 balls (result: 1 red, 1 black)
- 4 balls (result: 2 red, 2 black)
- 8 balls (result: 3 red, 5 black)
- 20 balls (result: 8 red, 12 black)

Randomness!

Example

Proportions of the colors in an urn (red or black balls)

- **Draw** Take a ball, record its color, and replace it.
- **Expectation** The proportion of red balls will converge toward the exact population proportion, as the number of draws increases.

Randomness!

Normal distribution

When a quantity is influenced by a large number of sources of variation, and these are all very small and independent of one another, it can be shown that the values of this quantity are distributed according to an analytical distribution known as the Laplace-Gauss law, also called the **normal distribution**.

Extrait de Vessereau A. La statistique, Que sais-je ?, PUF

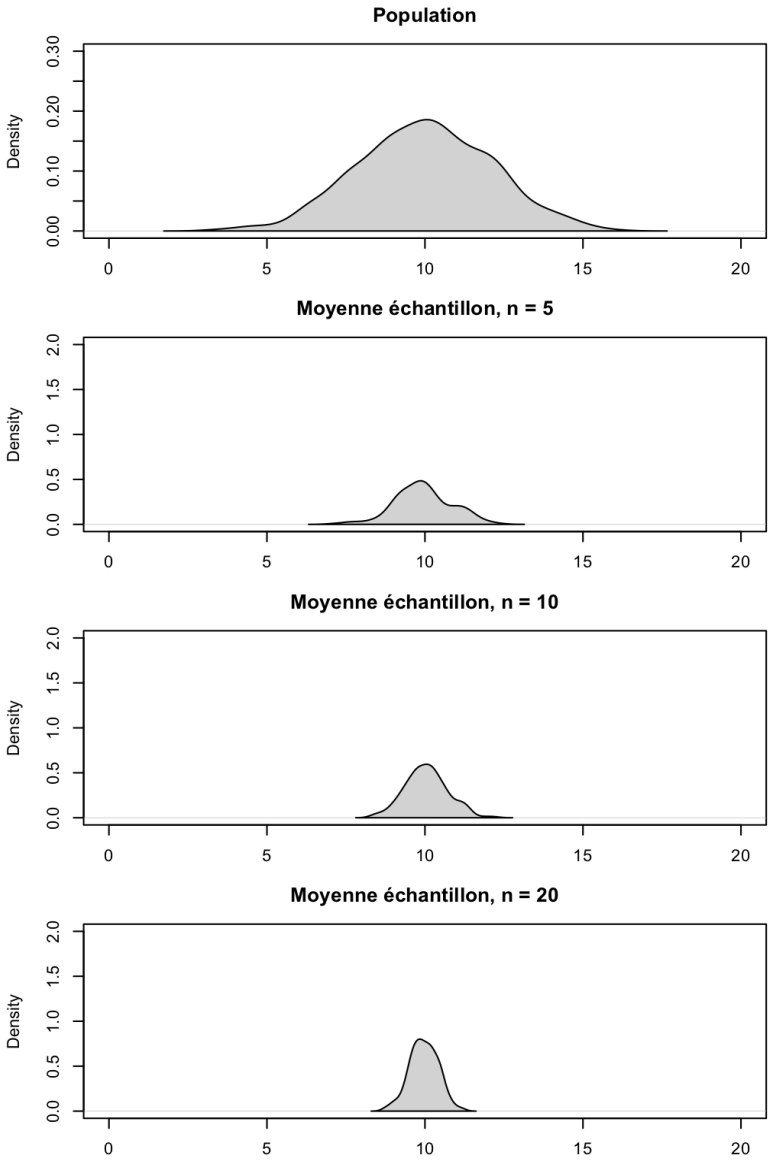
Randomness!

The law of large numbers ... or how the normal distribution can predict the behavior of a random variable as the number of observations becomes large.

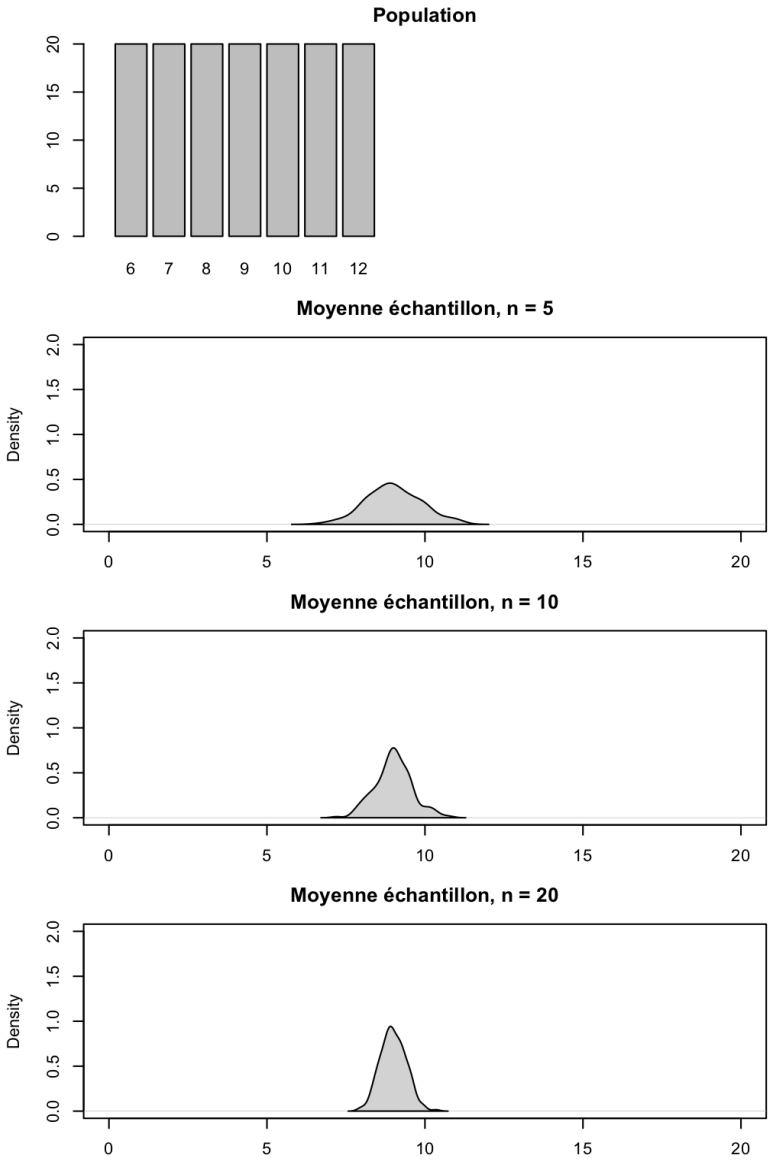
Case of the mean

- $$\hat{\mu} = \mu$$
- $$\hat{\sigma} = \sigma/\sqrt{n}$$

Randomness!


Population
Density
0.00
0.10
0.20
0.30
0
5
10
15
20
Moyenne échantillon, n = 5
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
Moyenne échantillon, n = 10
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
Moyenne échantillon, n = 20
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20

Randomness!


Population
0
5
10
15
20
6
7
8
9
10
11
12
Moyenne échantillon, n = 5
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
Moyenne échantillon, n = 10
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20
Moyenne échantillon, n = 20
Density
0.0
0.5
1.0
1.5
2.0
0
5
10
15
20

Estimation

We define:

$$\textit{Population } U \text{ with mean } \mu \text{ and variance } \sigma^2$$
$$\textit{Sample } S \text{ with mean } \hat{\mu} \text{ and variance } \hat{\sigma}^2$$
$$\textit{Variable } X_i, i \in S$$

The sample is a statistical series for which we can determine:

$$\text{the mean } \hat{\mu}_S = \frac{1}{n} \sum_{i \in S} X_i$$
$$\text{the variance } \hat{\sigma}^2_S = \frac{1}{n-1} \sum_{i \in S} (X_i - \hat{\mu}_S)^2$$

Estimation

The **mean** of the sample $\hat{\mu}_S$ is an estimator of the population mean:
$\hat{\mu}_S = \frac{1}{n} \sum_{i \in S} X_i$

We show that $E(\hat{\mu}_S) = \mu$
➔ $\hat{\mu}_S$ is an unbiased estimator of the mean μ of population U.

For the **variance**, it is more complicated:

$$\text{Sample variance } : \hat{\sigma^2}_S = \frac{1}{n-1} \sum_{i \in S} (X_i - \hat{\mu}_S)^2$$

$$\neq \text{population variance, } \sigma^2$$

Estimation

Definition: sampling without replacement

the observation selected at the i-th draw is not returned to the population before subsequent selections.

NB

Each draw reduces the population U by one unit.

➔ The observations are no longer random variables independent of one another.

Attention

Forest surveys are actually neither with nor without replacement.

Estimation

Consider simple random sampling **with replacement**.
The variance of the mean is $Var(\mu) = \frac{\sigma^2}{n}$.

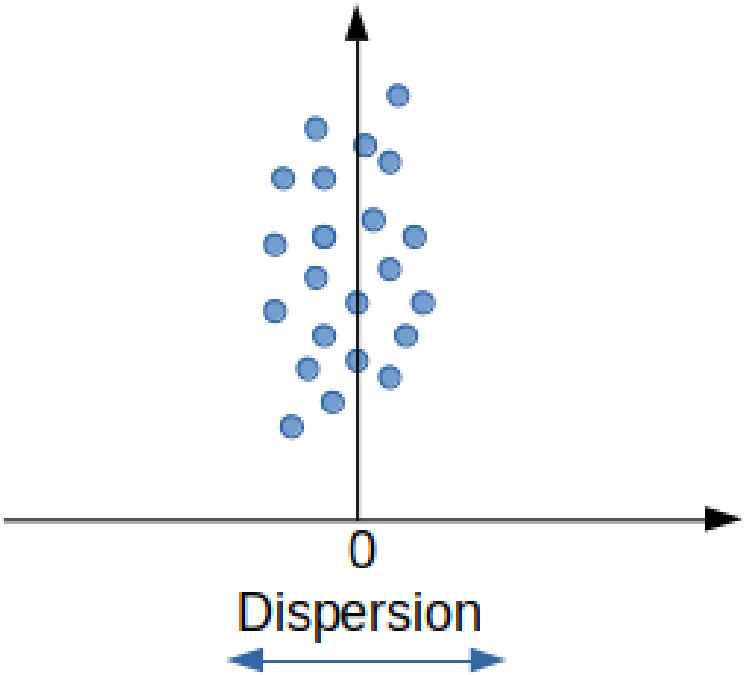
But we do not know σ^2, we only have the sample variance $\hat{\sigma}^2_S$, therefore

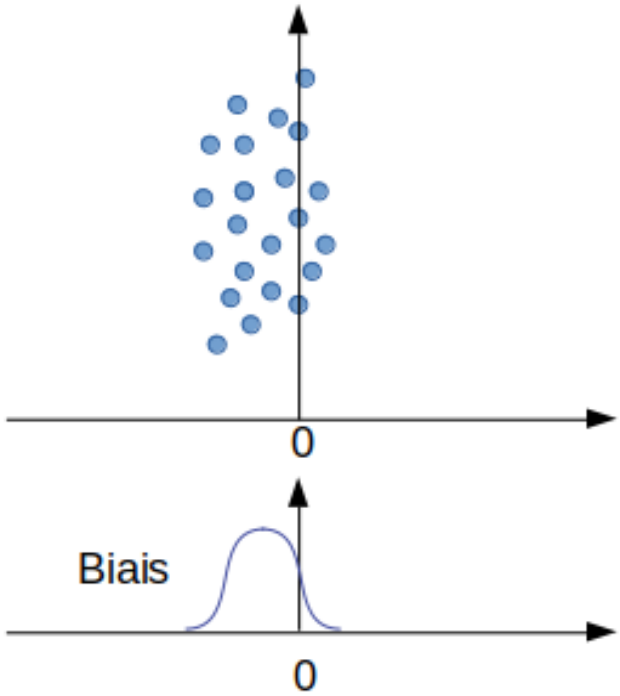
$$\begin{aligned} Var(\hat{\mu}_S) &= \frac{\hat{\sigma}^2_S}{n} \\ &= \frac{1}{n(n-1)} \sum_{i=1}^{n} \left(X_i - \hat{\mu}_S\right)^2 \end{aligned}$$

It can be shown that $\widehat{Var}(\hat{\mu}_S)$ is an unbiased estimator of $Var(\hat{\mu}_S)$.

Estimation

Dispersion / bias





Estimation

For **sampling without replacement**, the variance is written differently:

$$\begin{aligned}\widehat{Var}(\hat{\mu}_S) &= \frac{N-n}{N-1}\frac{\hat{\sigma}_S^2}{n-1} \\ &= \frac{N-n}{N-1}\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i-\hat{\mu}_S\right)^2\end{aligned}$$

which is often written differently by setting $f = \frac{n}{N}$:

$$\begin{aligned}\widehat{Var}(\hat{\mu}_S) &= (1-f)\frac{\hat{\sigma}_S^2}{n} \\ &= \frac{N-n}{N}\frac{1}{n(n-1)}\sum_{i\in S}\left(X_i-\hat{\mu}_S\right)^2\end{aligned}$$

Estimation

For large populations, $\lim_{n\to\infty}\left(\frac{N}{N-1}\right)=1$
therefore the estimate of the variance of $\hat{\mu}_S$ can be estimated as:

$$\widehat{Var}(\hat{\mu}_S)=\frac{(1-f)}{n-1}\sum_{i=1}^{n}\left(X_i-\hat{\mu}_S\right)^2$$

Likewise, when n is very small compared with N, $1-f$ approaches 1:

$$\widehat{Var}(\hat{\mu}_S)\approx\frac{1}{n-1}\sum_{i=1}^{n}\left(X_i-\hat{\mu}_S\right)^2$$

Finite population correction

In national forest inventories, these approximations are always assumed to hold.
Inventories for small domains cannot make this assumption.

Estimation of a total

Case of sampling with replacement

$$\hat{\tau}_X = N\,\hat{\mu}_S \quad \left(= \frac{N}{n}\sum_{i\in S} X_i\right)$$

with variance

$$Var\left(\hat{\tau}_X\right) = N^2\frac{\sigma^2}{n}$$

which is estimated by $\widehat{Var}\left(\hat{\tau}_X\right) = N^2\frac{s^2}{n}$

Estimation of a total

Case of sampling with replacement

$$\hat{\tau}_X = N\,\hat{\mu}_S$$

In practice, N is not always known and may not be defined.
We therefore use the mean density of X, expressed per *ha*
and N is replaced by the area of the study domain A, known without error:

$$\hat{\tau}_X = A\,\hat{\mu}_{Sha}$$

Estimation of a total

Case of sampling without replacement

$$\hat{\tau}_X = N\,\hat{\mu}_S$$

Thus, it is the same estimator as in the case of sampling with replacement. The variance estimator is somewhat more complex

$$\widehat{Var}\left(\hat{\tau}_X\right) = N^2(1-f)\frac{\hat{\sigma}_s^2}{n}$$

Let us return to stratified sampling
Estimation of a total

Each stratum h has its own estimate of the mean $\hat{\mu}_{S,h} = \frac{1}{n_h} \sum_{i \in (\text{S et h})} X_i$

The total of X in stratum h is therefore $\hat{\tau}_h = N_h \hat{\mu}_{S,h}$.

Since the strata are disjoint and exhaustive ($N = \sum_h N_h$), the total over the domain is:

$$\hat{\tau} = \sum_{h=1}^{H} \hat{\tau}_h$$

Stratified sampling

The population mean is estimated as:

$$\hat{\mu}_S = \frac{N_1\hat{\mu}_1 + N_2\hat{\mu}_2 + ... + N_H\hat{\mu}_H}{N}$$

Let W_h denote the weight of stratum h, defined as $W_h = \frac{N_h}{N}$, the mean can be written as:

$$\hat{\mu}_S = \sum_h W_h\hat{\mu}_{S,h}$$

Stratified sampling

$$\widehat{Var(\hat{\mu}_S)} = \sum_h^H W_h^2 \widehat{Var(\hat{\mu}_{S,h})}$$

where $\widehat{Var(\hat{\mu}_{S,h})}$ is the estimate of the variance in stratum h, namely:

$$\widehat{Var(\hat{\mu}_{S,h})} = \frac{N_h - n_h}{N_h} \frac{1}{n_h(n_h-1)} \sum_{i=1}^{n_h} \left(X_{h,i} - \hat{\mu}_h\right)^2$$

in the case of sampling without replacement.

Estimation

Estimation of the total:

$$\hat{\tau} = \sum_{h=1}^{H} N_h \bar{y}_h$$

with variance estimated as:

$$\widehat{Var}(\hat{T}_{st}) = \sum_{h=1}^{H} N_h^2 (1 - f_h) \frac{s_h^2}{n_h}$$

Unequal sampling probabilities

Until now, all units had the same probability of belonging to the sample.
But this is only a special case (obtained by construction).
Otherwise, the generalization of the previous estimators is the Horvitz-Thompson estimator

$$\hat{\tau} = \sum_i \frac{y_i}{\pi_i}$$

where π_i is the probability that unit i belongs to the sample.

Definition

π_i is the inclusion probability of unit i in the sample

The Horvitz-Thompson estimator

The Horvitz-Thompson theorem (1952) concerns samples drawn from a population $\mathcal{U}$ in order to estimate the total of an arbitrary function y:

$$T_y = \sum_{i \in \mathcal{U}} y_i$$

The theorem states that, if a sample S is drawn from $\mathcal{U}$ such that the probability π_i that point i is in S is positive for every $i \in \mathcal{U}$ then,

$$\hat{T}_y(S) = \sum_{i \in S} \frac{y_i}{\pi_i}$$

(Here $i \in S$ rather than $i \in \mathcal{U}$—notice the difference!)

Horvitz-Thomson, example.

Numerical example

Consider a population of 4 elements, from which 2 elements are sampled without replacement.

element	a	b	c	d
value	1	2	3	4

Estimation

Horvitz-Thomson, example.

The set of possible and equally likely samples contains 6 elements:

$$\begin{pmatrix} 1 & (a,b) \\ 2 & (a,c) \\ 3 & (a,d) \\ 4 & (b,c) \\ 5 & (b,d) \\ 6 & (c,d) \end{pmatrix}$$

Notice that each unit appears 3 times, and therefore has a selection probability of 0.5 ($\pi_i = 0.5$).

Estimation

Horvitz-Thomson, example.

The estimates obtained from these samples show that the mean and the total estimate are unbiased.

$$\mu = \frac{1}{4}\sum_{i=1}^{4} x_i = \frac{(1+2+3+4)}{4} = 2.5$$

Sample	pair	values	$\hat{\mu}$	total $\tau = 4 \times \hat{\mu}$
1	(a,b)	1,2	1.5	6
2	(a,c)	1,3	2	8
3	(a,d)	1,4	2.5	10
4	(b,c)	2,3	2.5	10
5	(b,d)	2,4	3	12
6	(c,d)	3,4	3.5	14
Mean		2.5	2.5	$\frac{1}{6}\sum \tau$ =10

Estimation

Horvitz-Thomson, example.

Suppose that sample (c, d) cannot be selected.

The sampling probabilities of the units become unequal:

unit	possible pairs	probability
a	(a,b), (a,c), (a,d)	3/5
b	(a,b), (b,c), (b,d)	3/5
c	(a,c), (b,c)	2/5
d	(a,d), (b,d)	2/5

Estimation

Horvitz-Thomson, example.

Sample	pair	values	$\hat{\mu}$	τ	$\tau_{HT} = \sum_{i=1}^{2} \frac{x_i}{\pi_i}$
1	(a,b)	1,2	1.5	6	$1.(5/3) + 2.(5/3) = 15/3$
2	(a,c)	1,3	2	8	$1.(5/3) + 3.(5/2) = 55/6$
3	(a,d)	1,4	2.5	10	$1.(5/3) + 4.(5/2) = 70/6$
4	(b,c)	2,3	2.5	10	$2.(5/3) + 3.(5/2) = 65/6$
5	(b,d)	2,4	3	12	$2.(5/3) + 4.(5/2) = 80/6$
6	(c,d)	3,4			
Mean		2.5	2.3	$46/5 = 9.2$	$300/6/5 = 10$

Estimation errors

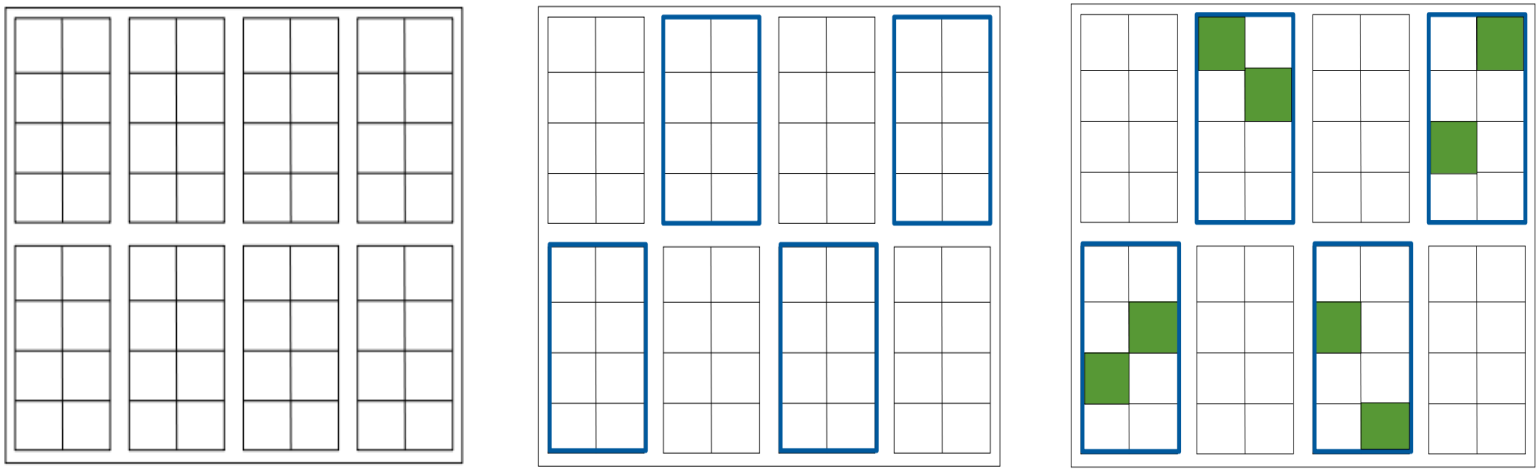
We almost always restrict ourselves to sampling errors.
Sampling errors are calculated at the 95% confidence level as

$$e.e.(\hat{\tau}@95\%) = \sqrt{\left(\widehat{Var}(\hat{\tau})\right)} \times t_{Student}(95\%)$$

Complex sampling

Multistage sampling

Two-stage designs



The first stage covers the entire population.
We assume that measuring all plots is unnecessary:
random selection of plots within the first-stage units

Definitions

PSU : Primary Sampling Unit
SSU : Secondary Sampling Unit

Two-stage designs

The total of Y for the population is calculated as:

$$\tau_Y = \sum_{i=1}^{N} \sum_{j=1}^{N_i} y_{ij}$$

where

- N is the number of primary units
- N_i is the number of secondary units in primary unit i
- y_{ij} the value of y in secondary unit j of i

Two-stage designs

Cluster sampling is a special case of two-stage sampling: in cluster sampling, all SSUs are selected for estimation.
Consequences for calculations (mean of means)

Two-phase sampling

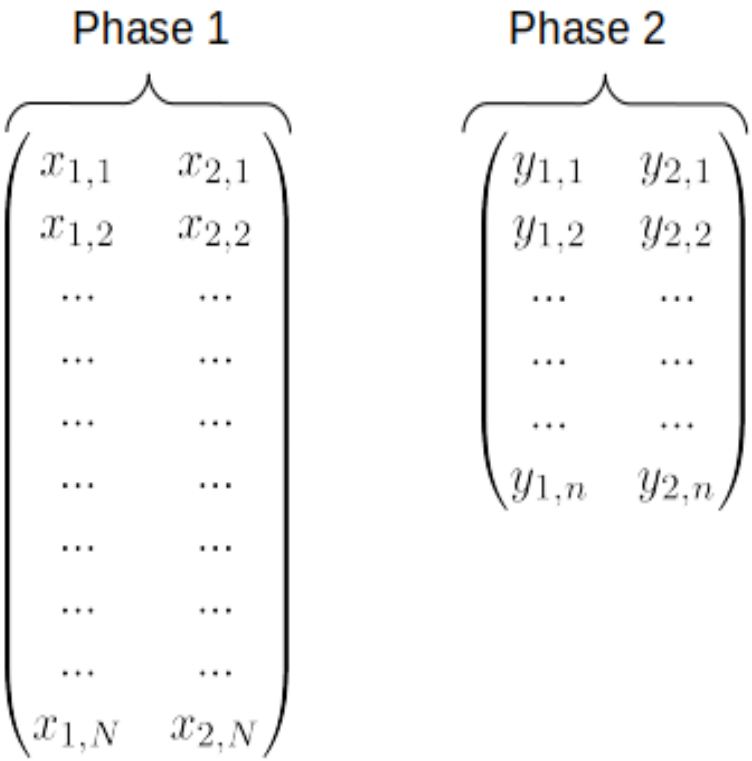
The characteristic of interest is not always the easiest to measure. Another characteristic, called an auxiliary characteristic, may be easier to measure.

NB

Stratification $\neq$ post-stratification

Multiphase sampling

Two-phase sampling



The principle is to have a large nombre de points de phase 1 ($X^{(1)}$) low-cost measurements
and a small number of points with observations or measurements of variables of interest ($Y^{(2)}$).

Multiphase sampling

Two-phase sampling

But the relationship between $X^{(1)}$ and $Y^{(2)}$ is sometimes weak.
In practice, in forest inventories, the first phase is photo-interpretation. The second phase consists of field measurements:

Phase 1	N	Partition into H strata
Phase 2	n « N	Mean within strata

Multiphase sampling

Two-phase sampling

Let A_D be the area of the study domain,
N_h the number of points in a given stratum h,
N the total number of first-phase points.

$$\tau_h = A_D \times \frac{N_h}{N} \times \hat{\mu}_h$$

This is indeed the stratified estimation formula.

Multiphase sampling

Reminder: in stratified sampling, the stratum weights are known:
W_h is defined as $W_h = \frac{N_h}{N}$
the mean can therefore be written as:

$$\hat{\mu}_S = \sum_h W_h \hat{\mu}_{S,h}$$

Here the stratum sizes are known.

$\neq$

In post-stratified sampling, the stratum size is **estimated** by the sampling process itself, and therefore has an associated estimation error.
The area of stratum A_h is in fact estimated as

$$\hat{A}_h = A_D \times \frac{N_h}{N}$$

Two-phase sampling

The error calculation must account for the two sources of error corresponding to each phase.
The variance is calculated using conditioning on n_h.

$$Var(\hat{\mu}_{post}) = Var[E(\hat{\mu}_{post}|n_h)] + E[Var(\hat{\mu}_{post}|n_h)]$$
$$\approx \frac{(1-f)}{n}\sum_{h=1}^{H}\frac{N_h}{N}\hat{\sigma}_h^2 + \frac{(1-f)}{n^2}\sum_{h=1}^{H}\frac{N-N_h}{N}\hat{\sigma}_h^2$$

Summary formulas (discrete populations)

Simple random sampling with replacement

Let U be a population of size N, and let S be a sample of n draws **with replacement**.

$$\bar{y}_s = \frac{1}{n}\sum_{i\in s} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i\in s}(y_i - \bar{y}_s)^2.$$

	Mean	$\hat{\bar{Y}} = \bar{y}_s$
	Total	$\hat{T} = N\bar{y}_s = \frac{N}{n}\sum_{i\in s} y_i$
Sampling variance		$\widehat{Var}(\hat{\bar{Y}}) = \frac{s^2}{n}$
		$\widehat{Var}(\hat{T}) = N^2\frac{s^2}{n}$

Simple random sampling without replacement (SRSWOR)

Let U be a population of size N, and let S be a sample of n distinct units. Define

$$\bar{y}_s = \frac{1}{n}\sum_{i\in s} y_i, \qquad s^2 = \frac{1}{n-1}\sum_{i\in s}(y_i - \bar{y}_s)^2, \qquad f = \frac{n}{N}.$$

	Mean	$\hat{\bar{Y}} = \bar{y}_s$
	Total	$\hat{T} = N\bar{y}_s = \frac{N}{n}\sum_{i\in s} y_i$
Sampling variance		$\widehat{Var}(\hat{\bar{Y}}) = (1-f)\frac{s^2}{n}$
		$\widehat{Var}(\hat{T}) = N^2(1-f)\frac{s^2}{n}$

Finite population correction

The factor $(1-f)$ accounts for sampling without replacement.

Stratified sampling

The population is divided into H exhaustive and mutually exclusive strata. In stratum h:

$$N_h = \text{population size}, \qquad n_h = \text{sample size}, \qquad W_h = \frac{N_h}{N}, \qquad f_h = \frac{n_h}{N_h}.$$

Let $\bar{y}_h$ and s_h^2 be the sample mean and sample variance within stratum h.

Mean	$\hat{\bar{Y}}_{st} = \sum_{h=1}^{H} W_h \bar{y}_h$
Total	$\hat{T}_{st} = \sum_{h=1}^{H} N_h \bar{y}_h$
Sampling variance	$\widehat{Var}(\hat{\bar{Y}}_{st}) = \sum_{h=1}^{H} W_h^2 (1 - f_h) \frac{s_h^2}{n_h}$
	$\widehat{Var}(\hat{T}_{st}) = \sum_{h=1}^{H} N_h^2 (1 - f_h) \frac{s_h^2}{n_h}$

Unequal inclusion probabilities: Horvitz–Thompson

Each population unit i has a known, positive inclusion probability

$$\pi_i = P(i \in S) > 0.$$

The corresponding expansion weight is

$$w_i = \frac{1}{\pi_i}$$

Total	$\hat{T}_{HT} = \sum_{i \in S} \frac{y_i}{\pi_i} = \sum_{i \in S} w_i y_i$
Mean	$\hat{\bar{Y}}_{HT} = \frac{\hat{T}_{HT}}{N} = \frac{1}{N} \sum_{i \in S} \frac{y_i}{\pi_i}$

Unequal inclusion probabilities: Horvitz–Thompson

Let $\pi_{ij} = P(i, j \in S)$ be the joint inclusion probability. Then

$$\widehat{Var}(\hat{T}_{HT}) = \sum_{i \in U} \sum_{j \in U} \frac{\pi_{ij} - \pi_i \pi_j}{\pi_{ij}} \frac{y_i}{\pi_i} \frac{y_j}{\pi_j} \widehat{Var}(\hat{\bar{Y}}_{HT})$$

$$\widehat{Var}(\hat{\bar{Y}}_{HT}) = \frac{1}{N^2} \widehat{Var}(\hat{T}_{HT})$$

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